



SbYS1 and SbWRKY72 regulate Cd tolerance and accumulation in sweet sorghum

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Abstract

Main conclusion SbYS1 and its upstream transcription factor SbWRKY72 were involved in Cd tolerance and accumulation and are valuable for developing sweet sorghum germplasm with high-Cd tolerance or accumulation ability through genetic manipulation.

Abstract Cadmium (Cd) is highly toxic and can severely affect human health. Sweet sorghum, as an energy crop, shows great potential in extracting cadmium from Cd-contaminated soils. However, its molecular mechanisms of Cd-tolerance and -accumulation remain largely unknown. Here, we isolated a YSL family gene *SbYS1* from the sweet sorghum genotype with high Cd accumulation ability and the expression of *SbYS1* in roots was induced by cadmium. GUS staining experiment exhibited that *SbYS1* was expressed in the epidermis and parenchyma tissues of roots. Further subcellular localization analysis suggested that *SbYS1* was localized in the endoplasmic reticulum and plasma membrane. Yeast transformed with *SbYS1* exhibited a sensitive phenotype compared to the control when exposed to Cd-NA (chelates of cadmium and nicotianamine), indicating that *SbYS1* may absorb cadmium in the form of Cd-NA. Arabidopsis overexpressing *SbYS1* had a longer root length and accumulated less Cd in roots and shoots. *SbWRKY72* bound to the promoter of *SbYS1* and negatively regulated the expression of *SbYS1*. Transgenic Arabidopsis of *SbWRKY72* showed higher sensitivity to cadmium and increased cadmium accumulation in roots. Our results provide references for improving the phytoremediation efficiency of sweet sorghum by genetic manipulation in the future.

Keywords Cadmium · SbWRKY72 · *SbYS1* · Sweet sorghum

Abbreviations

DMA Deoxymugineic acid
IRT Iron-regulated transporter

NA Nicotianamine
YSL Yellow Stripe-Like

Introduction

Cadmium, as a nonessential element, has adverse effects on plants and humans (Liu et al. 2017; Zhang et al. 2023). In China, rapid industrialization over the past 30 years has led to widespread cadmium contamination in agricultural soils (Wang et al. 2019). Plants can absorb cadmium in soil via the apoplastic pathway or by transporters for the uptake of some metal elements that are essential for plant growth and development, such as calcium, iron and zinc (Thomine et al. 2000; Perfus-Barbeoch et al. 2002; Lux et al. 2011; Maathuis and Diatloff 2013; Menguer et al. 2013; Feng et al. 2018). A proportion of cadmium is accumulated in edible parts of crops or vegetables which can be taken up by humans through the food chain, and severely threaten human health

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(Schaefer et al. 2020). Therefore, it is essential to remediate cadmium-polluted soils for food security. Cadmium can be extracted by physical and chemical approaches (Tang et al. 2016; Ziarati et al. 2023). However, these approaches usually cause damage to soil structure or even secondary pollution, and most of them are costly (Khairy et al. 2014; Rajendran et al. 2022). Phytoremediation is an economical and environment-friendly remediation technology, which can extract and recycle heavy metals through plants, especially some hyperaccumulators (Doty 2008). However, most hyperaccumulators present a slow growth rate, accompanied by a low biomass and small economic profit, which restricts the application of these plants in phytoremediation (Liu et al. 2011). Notably, some energy plants with high biomass have been proposed for the phytoremediation of heavy metal-contaminated soils, such as Napier grass (*Pennisetum purpureum* Schumacher), switchgrass (*Panicum virgatum* L.) and sorghum (*Sorghum bicolor* (L.) Moench) (Metwali et al. 2013; Wang et al. 2015; Chen et al. 2017).

Sweet sorghum is a variety of sorghum and has high photosynthesis efficiency as a C₄ plant, which has been considered as a promising energy plant (Han et al. 2012; Jia et al. 2017). Previous studies have reported that sorghum or sweet sorghum can extract heavy metals from soil or water (Pinto et al. 2004; Bonfranceschi et al. 2009; Metwali et al. 2013; Salman et al. 2013; Al Chami et al. 2015). For instance, sorghum showed higher tolerance to cadmium and copper stresses than wheat and maize, and could remove lead, cadmium, and copper from water effectively (Metwali et al. 2013; Salman et al. 2013).

Using sweet sorghum for the remediation of Cd-contaminated soils not only provides raw materials for bioenergy production but also extracts heavy metals from the contaminated land simultaneously. The fibrous residues left after bioenergy production can be further burned to generate electricity or process heat, and cadmium can be further recycled from the ash (Woods 2001; Li 2013). Through this process, cadmium is transferred from the food chain into the energy chain, which avoids adverse effects on human health (Jia et al. 2017). However, previous studies suggested that the ability of sweet sorghum to absorb cadmium is limited and most cadmium is accumulated in roots (Soudek et al. 2014; Jia et al. 2016, 2017), so promoting cadmium uptake and translocation is crucial for improving the phytoremediation capacity of sweet sorghum.

Heavy metal transporters have played important roles in Cd uptake and translocation (Connolly et al. 2002; Lanquar et al. 2005; Morel et al. 2009; Wang et al. 2013). In our previous study, one sweet sorghum genotype, H18 with high Cd-accumulation and -tolerance abilities, was identified through germplasm screening (Jia et al. 2017; Feng et al. 2018). Further transcriptome analysis showed that the relative expression level of a transporter gene *Sobic.004G299500* in H18

increased more than 40 times under cadmium stress compared to control, indicating *Sobic.004G299500* may function in cadmium uptake and tolerance of sweet sorghum (Feng et al. 2018). *Sobic.004G299500* (named *SbYS1*) encodes a Yellow Stripe-Like (YSL) transporter belonging to the oligopeptide transporter (OPT) family, which has been found to be involved in the uptake and long-distance transport of several metals (Conte and Walker 2012). ZmYS1 from maize was the first identified YSL transporter with the ability of transporting chelates of Fe, Ni, Zn, Cu, Mn and Cd with deoxymugineic acid (DMA), mugineic acid (MA) or nicotianamine (NA) (Curie et al. 2001; Schaaf et al. 2004). Wang et al. (2013) reported that the expression of *BjYSL7* was induced in both roots and shoots of *Brassica juncea* under Fe, Ni and Cd stress, and transgenic *BjYSL7* tobacco exhibited elevated tolerance to Ni and Cd, and their concentrations in shoots increased obviously compared to wild-type. Feng et al. (2017) reported that *SnYSL3* can transport complexes of Fe(II), Cu, Zn and Cd with NA, and expressing *SnYSL3* in Arabidopsis increased the translocation capacity of Fe and Cd from root to shoot under Cd stress. *HvYS1* from barley (*Hordeum vulgare*), whose expression is induced by iron deficiency, encoded a Fe-selective metal transporter that mainly expressed in the root epidermal cells, and rice expressing *HvYS1* reduced Cd uptake in roots and leaves, root-to-shoot translocation, and accumulation in the seeds (Murata et al. 2006; Ueno et al. 2009; Banakar et al. 2017).

Transcription factors of the WRKY family are characterized with a conserved WRKY domain of 60 amino acids region and were identified to be involved in plants resistance to heavy-metal stresses (Eulgem et al. 2000; Sheng et al. 2019; Mirza et al. 2022). Yang et al. (2016) showed that *ThWRKY7* from *Tamarix hispida* may regulate the expression of a V-ATPase c subunit gene *ThVHAc1* and further affect plant tolerance to cadmium stress. Sheng et al. (2019) reported that *WRKY13*, a WRKY transcription factor from Arabidopsis, can activate the expression of Cd²⁺ extrusion gene *PDR8* to positively regulate Cd tolerance. Zhang et al. (2023) found that *AtWRKY33* positively regulated Cd tolerance by triggering the transcription of *Arabidopsis Tóxicos en Levadura 31 (ATL31)*, which encoded an E3 ligase that affected the iron-regulated transporter (IRT1) protein levels by ubiquitin-conjugated degradation. *WRKY72* is one of the WRKY members that has been reported to be involved in the basal immunity of tomato, Arabidopsis and rice (Bhattarai et al. 2010; Hou et al. 2019). However, whether *WRKY72* functions in plant's response to heavy-metal stress remains unclear.

Although the YSL family transporter *SbYS1* was significantly upregulated by cadmium stress, whether *SbYS1* is involved in cadmium uptake and translocation is unclear, and the underlying regulatory mechanisms need to be further revealed. Here, we characterized *SbYS1* and its upstream

transcription factor SbWRKY72 and identified both of them functioning in Cd tolerance and accumulation. This study can provide references for improving the remediation efficiency of sweet sorghum in Cd-contaminated soils.

Material and methods

Plant materials and growth conditions

Sweet sorghum genotype ‘UMM EL TEIMAN’ was screened from ninety-six germplasm obtained from USDA (United States Department of Agriculture) for its high ability of Cd accumulation and tolerance (Jia et al. 2017; Feng et al. 2018), and its accession number is PI 152873 in Plant Genetic Resources Conservation Unit of USDA (<https://npgsweb.ars-grin.gov/gringlobal/search>).

The experimental design and growth conditions of sweet sorghum referred to the methods described by Jia et al. (2016, 2017). In brief, seedlings with uniform growth were selected and cultivated hydroponically with Hoagland solution, Cd (as CdCl₂) was added to containers as the experimental needs. To ensure that sweet sorghum was in the same biological age and growth rhythm, seedlings treated with 10 μM CdCl₂ for 1 h, 6 h, 12 h, 24 h and 3 d were harvested together with control samples at the same time according to the methods mentioned by Fan et al. (2013). For phenotype analysis, the transgenic Arabidopsis seedlings of *SbYS1* and *SbWRKY72* were grown on 1/2 MS media with the corresponding concentration of 40 and 75 μM CdCl₂ for 2 weeks, then the roots and shoots were harvested separately.

Cloning of *SbYS1* and *SbWRKY72*

Roots of sweet sorghum exposed to 10 μM CdCl₂ for 24 h were used to extract total RNA with the RNeasy Plant Mini Kit (TransGen Biotech), and then the total RNA was converted to cDNA using the SuperScript II Reverse Transcriptase (Invitrogen). The full-length cDNA of *SbYS1* was amplified by the PCR using primers 5'-CAGCGATCCAGCGAAGCC-3' and 5'-CTCTCGGAGAAGGAGAACCCAC-3' while the basic PCR primers for molecular cloning *SbWRKY72* are 5'-TGAGCCAGAAATAAGTAGCAGACG-3' and 5'-CAGTCTCTATCCCTGCCCTACAAT-3'. The entire *SbYS1* and *SbWRKY72* cDNA from sweet sorghum was subcloned into the corresponding vectors as necessary.

Phylogenetic analysis

Sequence alignment was conducted by BioEdit software (Hall 1999). The phylogenetic tree was constructed by MEGA7 (Kumar et al. 2016).

GUS assay

The *SbYS1* promoter, which contained 1307 bp sequence upstream of the *SbYS1* initiating ATG, was cloned and fused to the GUS reporter gene to explore its tissue expression pattern. The *SbYS1* promoter-GUS construct was transformed into Arabidopsis using the floral dip method (Clough and Bent 1998). T2 seeds were germinated on 1/2 MS medium containing hygromycin to select for transformants, then transferred transgenic lines to a new medium adding 20 μM CdCl₂ for three days. Histochemical GUS staining was conducted with reference to the method mentioned by Jefferson et al. (1987). Sections of 4 μm thickness were obtained as described by Jia et al. (2016), and were observed with a light microscope and photographed with a CCD camera.

The expression level of *SbYS1* and *SbWRKY72*

The expression level of *SbYS1* and *SbWRKY72* was measured by RT-qPCR with a Stratagene Mx3000P instrument (Stratagene, Santa Clara, CA, USA). The relative quantification method ($2^{-\Delta\Delta C_T}$) was used to evaluate the expression of target genes relative to β-actin gene in sorghum or *AtActin* gene in transgenic Arabidopsis (Livak and Schmittgen 2001). The primers used in RT-qPCR analysis are listed in Table S1.

Antibodies and immunoblotting assay

Anti-WRKY72 antibodies were obtained in rabbits through WUHAN GENECREATE BIOLOGICAL ENGINEERING CO., Ltd. Total proteins were extracted from the samples as described by Wang et al. (2018). After separation by SDS-PAGE, protein was transferred to a PVDF membrane for Western-blot analysis. ECL system was used to detect the secondary antibodies (goat anti-rabbit IgG coupled with horseradish peroxidase).

Subcellular localization of *SbYS1* and *SbWRKY72*

The subcellular localization of *SbYS1* and *SbWRKY72* was investigated through the tobacco instant expression system for its high reproducibility and reliability (Sparkes et al. 2006; Marion et al. 2008; Zhang and He 2021). Both of their cDNAs were introduced into pCAMBIA1300-163 vector, respectively, under the control of the CaMV 35S promoter. The intact leaves of one-month-old tobacco (*Nicotiana benthamiana*) plants were infiltrated with *Agrobacterium tumefaciens* strain GV3101 harboring pCAMBIA1300-*SbYS1*/*SbWRKY72*-GFP. RFP-HDEL was used as a marker of endoplasmic reticulum (ER). FM4-64 was used to stain the plasma membrane and H2B protein was used as a nucleus marker. Localization of fluorescent protein

was monitored using a confocal laser scanning microscope (SbYS1 for Leica TCS SP5, SbWRKY72 for Olympus FV1000MPE).

Functional complementation in yeast

The pYES2 vectors containing *SbYS1*, *AtIRT1*, *ZmYS1* and empty vector were transformed into *Saccharomyces cerevisiae* strain DY1457 (*MAT α ade6 can1 his3 leu2 trp1 ura3*) referred to the method by Gietz et al. (1992). Cd-DMA (the chelate of cadmium and deoxymugineic acid) or Cd-NA (the chelate of cadmium and nicotianamine) was obtained as the methods described in Schaaf et al. (2004) and Meda et al. (2007), then added to the medium. The transformed yeast cells were cultured in a synthetic dropout medium without uracil (SD-U) overnight. SD-U medium includes 0.67% yeast nitrogen base, 0.077% -ura drop-out supplement and 2% glucose. After being enriched and washed three times with sterile water, the yeast transformants were diluted to an initial OD value of 1.0 at 600 nm and three series of 1:10 dilutions with sterilized distilled water, five microliters of the cell suspension were spotted onto plates with SD-U medium that contained cadmium and 2% galactose. After being incubated at 30 °C for three days, the plates were photographed.

Overexpressing *SbYS1* and *SbWRKY72* in Arabidopsis

Verification of *SbYS1* and *SbWRKY72* function was conducted by heterologous expression in Arabidopsis due to the difficulty of genetic transformation in sweet sorghum (Raghuwanshi and Birch 2010; Ahmed et al. 2018). The full length of *SbYS1* and *SbWRKY72* cDNAs was fused with the pSN1301 vector and transformed into Arabidopsis (Columbia ecotype) according to the methods by Clough and Bent (1998). Two homozygous lines of transgenic plants in the T2 generation were used for the next experiment. Four-day-old seedlings were transplanted to 1/2 MS solid medium including 40 or 75 μM CdCl_2 . A week later, the root length and fresh weight of the shoots was measured. The chlorophyll content was measured according to the methods described by Jia et al. (2016) and Lichtenthaler (1987). The roots and shoots were harvested separately and dried at 70 °C until constant weight for the next experiment.

Yeast one-hybrid assay

Yeast one-hybrid assay is an in vitro method to analyze the intracellular interaction between DNA and proteins, which has been widely used in the interactive research of transcription factor and their target genes (Wanke and Harter 2009; Dey et al. 2012). In our study, yeast one-hybrid assay was introduced to study the regulation of transcription factor

SbWRKY72 on *SbYS1*. The *SbWRKY72* cDNA and promoter of *SbYS1* were separately inserted into the pB42AD and pLacZi2 μ vectors. Then the two integrated vectors were transformed into the yeast strain EGY48 simultaneously according to the Yeast Protocols Handbook (Clontech). After being cultured on SD/-Ura/- Trp at 30 °C for 4–6 days, transformed yeast was photographed.

Transcriptional activity assay in yeast

The coding sequences of full-length *SbWRKY72* were integrated into the pGBKT7 vector. The pGBKT7-*SbWRKY72* and the empty vector pGBTK7 were transformed into the yeast strain Y2H Gold together with the pGADT7 vector, respectively. The transformed yeast was selected on SD medium without tryptophan (Trp) and leucine (Leu); then the yeast transformant from SD/-Trp/-Leu was cultured on SD/-Trp/-Leu/-His (histidine) /-Ade (adenine) at 30 °C for 3 d.

Electrophoretic mobility shift assay (EMSA)

For the purpose of producing the recombinant His-*SbWRKY72* protein, the coding sequence (CDS) of *SbWRKY72* was introduced into the pCold TF vector and the fused vector was transformed into *Escherichia coli* (BL21). 1 mM isopropyl- β -D-1-thiogalactopyranoside (IPTG) was used to induce the recombinant His-*SbWRKY72* protein at 16 °C for 16–20 h. The recombinant protein was then purified using nickel affinity chromatography in accordance with the manufacturer's instructions. LightShift Chemiluminescent EMSA kit (Thermo Fisher Scientific, Waltham, MA USA) was used to carry out the EMSA assay as the manufacturer's instructions. Based on DNA fragments, two probes that contained the W-box and altered probes were created by Shanghai Sangon Biotechnology in China using the promoter sequence of *SbYS1*.

Dual luciferase (LUC) reporter assay

The *SbYS1* promoter region (1307 bp) was cloned into the pGreen II-0800-LUC vector as the reporter while the open reading frame (ORF) of *SbWRKY72* was introduced into pCAMBIA1300 as the effector. The *A. tumefaciens* strain GV3101 transformed with the pCAMBIA1300-*SbWRKY72*-GFP or pCAMBIA1300-GFP plasmids was co-infiltrated with GV3101 transformed with the pGreen II-0800-Pro*SbYS1*-LUC vector into tobacco (*N. benthamiana*) plants, respectively.

After 48 h of incubation at 22 °C, infiltrated plants were imaged using a digital CCD camera. Before luminescence testing, leaves were sprayed with 100 mM luciferin and left in the dark for 3–5 min. A minimum of five independent

LUC quantifications were evaluated, and the outcomes were consistent. A ratio of LUC to REN was used to express the promoter activity.

Cadmium determination

The dry samples were ground into powder and digested in 1 mL nitric acid at 200 °C for 4 h, then Cd concentrations were measured by inductively coupled plasma atomic emission spectrometry (ICP-AES, Thermo Fisher Scientific).

Genes' information

Genes Identifier of sorghum given in Phytozome 13 (<https://phytozome-next.jgi.doe.gov/>): *SbYS1*, *Sobic.004G299500*; *SbWRKY72*, *Sobic.005G117400*.

Statistical analysis

The statistical analysis was conducted by the SPSS 17.0 statistical package (SPSS Inc., Chicago, IL, USA). Differences were tested by Student's *t* test unless otherwise specified, and the values were considered statistically significant at $P < 0.05$.

Results

Isolation and phylogenetic analysis of *SbYS1*

The cDNA of *SbYS1* was cloned from sweet sorghum with a length of 2031 bp, encoding a protein consisting of 676 amino acids. Amino acid sequence analysis suggested that *SbYS1* exhibited high identity with *MsYSL1* (95.87%) of switchgrass and *ZmYS1* (92.38%) of maize (Fig. S1). According to the results of phylogenetic analysis, Yellow Stripe-Like proteins from Arabidopsis, rice, barley as well as other reported members can be divided into four groups obviously. *SbYS1* belongs to a subfamily including *MsYSL1*, *ZmYS1*, *OsYSL15*, *HvYS1*, *AtYSL1* and so on (Fig. S2).

Expression patterns of *SbYS1*

In the presence of Cd, the expression level of *SbYS1* was greatly induced in roots and increased first, then decreased at 24 h but remained higher than in controls (Fig. 1a). By contrast, the expression level of *SbYS1* in shoots remained at a low level except a slightly increase at 12 h (Fig. 1a). To further investigate the tissue expression patterns of *SbYS1*, we generated transgenic Arabidopsis plants carrying the 1307 bp *SbYS1* promoter region that was fused to *GUS* gene. Cross-sections observation exhibited that *GUS* activity

driven by *SbYS1* promoter was detected in the epidermis and parenchyma of roots, in the epidermis, parenchyma and stele of stems as well as in the vascular bundles and palisade tissue of leaves (Fig. 1b–d).

Subcellular localization of *SbYS1*

To examine the subcellular localization of *SbYS1*, the open reading frame (ORF) of *SbYS1* without termination codon were fused with GFP and transiently expressed in tobacco leaves. The green fluorescent signal was detected at the endoplasmic reticulum and coincident with the red signal from the ER marker RFP-HDEL (Fig. 2a). Besides, the green fluorescent signal merged with the red signal from the plasma membrane dye FM4-64 (Fig. 2b). These results suggest that *SbYS1* is simultaneously located at the endoplasmic reticulum and plasma membrane.

Transport activity assay of *SbYS1* in yeast

To investigate the function of *SbYS1*, the transport activity assay was carried out in a yeast expression system. *SbYS1*, *AtIRT1* and *ZmYS1* were inserted into pYES2 vectors and were transformed into the wild-type yeast DY1457, yeast strains transformed with empty vector, *AtIRT1* and *ZmYS1* were taken as control. *AtIRT1* can transport Cd^{2+} while *ZmYS1* can transport Cd-DMA (the chelate of cadmium and deoxymugineic acid) (Rogers et al. 2000; Schaaf et al. 2004). Under the control condition, there was no obvious difference among yeast strains containing *SbYS1*, *AtIRT1*, *ZmYS1* or empty vector (Fig. 3). In the presence of 20 μM CdCl_2 , the yeast expressing *AtIRT1* gene was unable to grow, while yeasts transformed with *SbYS1* and *ZmYS1* showed no significant difference compared to control (Fig. 3). When exposed to 20 μM CdCl_2 plus 10 μM DMA, these yeast strains exhibited the similar phenotype as under 20 μM CdCl_2 treatment. However, under the treatment of 20 μM Cd plus 10 μM NA, yeast strains containing *SbYS1* were more sensitive than yeast strains containing *ZmYS1* and empty vector (Fig. 3). These results suggested *SbYS1* may absorb cadmium in the form of Cd-NA chelate in sweet sorghum.

Overexpressing *SbYS1* in Arabidopsis enhanced its tolerance to cadmium

To further evaluate the role of *SbYS1* in plants, we expressed *SbYS1* in Arabidopsis and investigated their phenotype under control and Cd treatment. Under control conditions, there was no significant difference in root length, fresh weight and chlorophyll content between *SbYS1* transgenic lines and wild-type (Fig. 4a and b–d). In the presence of Cd, the root length of all transgenic lines was longer than that of wild-type (Fig. 4a and b), however, shoot fresh weight (Fig. 4c)

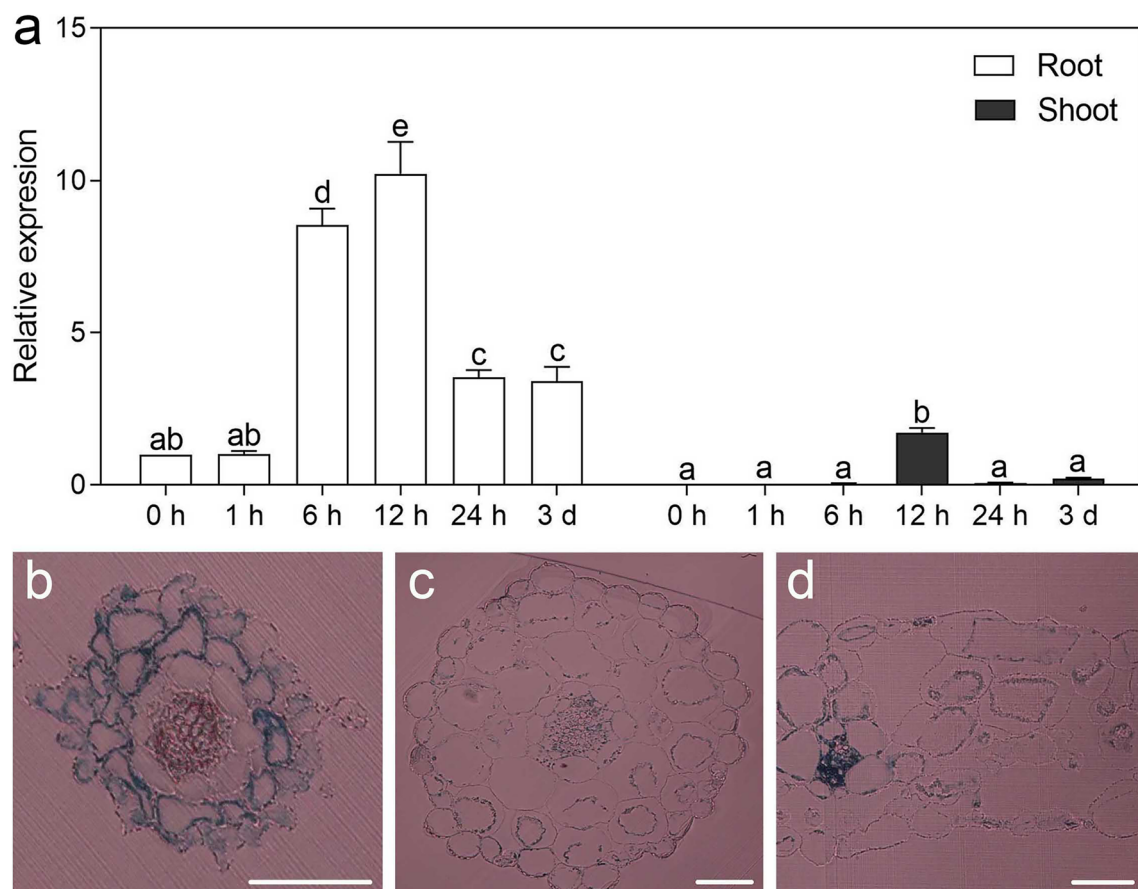


Fig. 1 The expression pattern and tissue-specific localization of *SbYS1*. **a** The expression level of *SbYS1* under 10 μM CdCl_2 treatment for 0, 1, 6, 12, 24 h and 3 d. Values are mean $\pm$ SE ($n=4$). **b-d**

Transverse sections of Arabidopsis expressing *GUS* driven by the *SbYS1* promoter under 20 μM CdCl_2 treatment for 3 d. **b** Roots, **c** Stems, **d** Leaves. Scale bars = 20 μm

and chlorophyll content (Fig. 4d) showed no significant difference between transgenic lines and wild-type. Unexpectedly, the accumulation of Cd decreased in shoots (Fig. 4e) and roots (Fig. 4f) of *SbYS1* transgenic lines. In addition, Cd translocation factors of transgenic lines were higher than that of wild-type although there was no significant difference (Fig. 4g).

SbYS1 was regulated by SbWRKY72

Plant transcription factor database PlantTFDB (<http://planttfdb.gao-lab.org/>) was used to predict the potential transcription factors that regulate the expression of *SbYS1* based on the promoter sequence of *SbYS1* and the sorghum genome. The results showed that there were 85 transcription factors that might bind to the *SbYS1* promoter (Table S2). According to the GO annotation information of these transcription factors in the PlantTFDB database, three transcription factors, Sobic.005G117400, Sobic.001G469100 and Sobic.003G370700, which are predicted to be related to the binding of metal ions or the regulation of nutrients,

were selected for further verification. Yeast one-hybrid experiments identified that only the yeast strains transformed with Sobic.005G117400 and *SbYS1* promoter turned blue (Fig. 5a), indicating Sobic.005G117400 combined with *SbYS1* promoter. *Sobic.005G117400* encodes a WRKY family transcription factor in sweet sorghum. We named Sobic.005G117400 as SbWRKY72 according to its annotation. Moreover, EMSA revealed that SbWRKY72 directly bound to the DNA fragments containing W-box elements (Fig. 5b). The transient expression analysis in *N. benthamiana* with luciferase reporter assay exhibited that SbWRKY72 repressed the promoter activity of *SbYS1* (Fig. 5c).

Characterization of SbWRKY72

Phylogenetic analysis showed that SbWRKY72 shared high similarities with ZmWRKY24 (78.79%), ZmWRKY75 (77.59%), and ZmWRKY45 (72.84%) at amino acid sequence level (Fig. S3). Subcellular localization showed that SbWRKY72 was located in the nucleus (Fig. 6a). Yeast

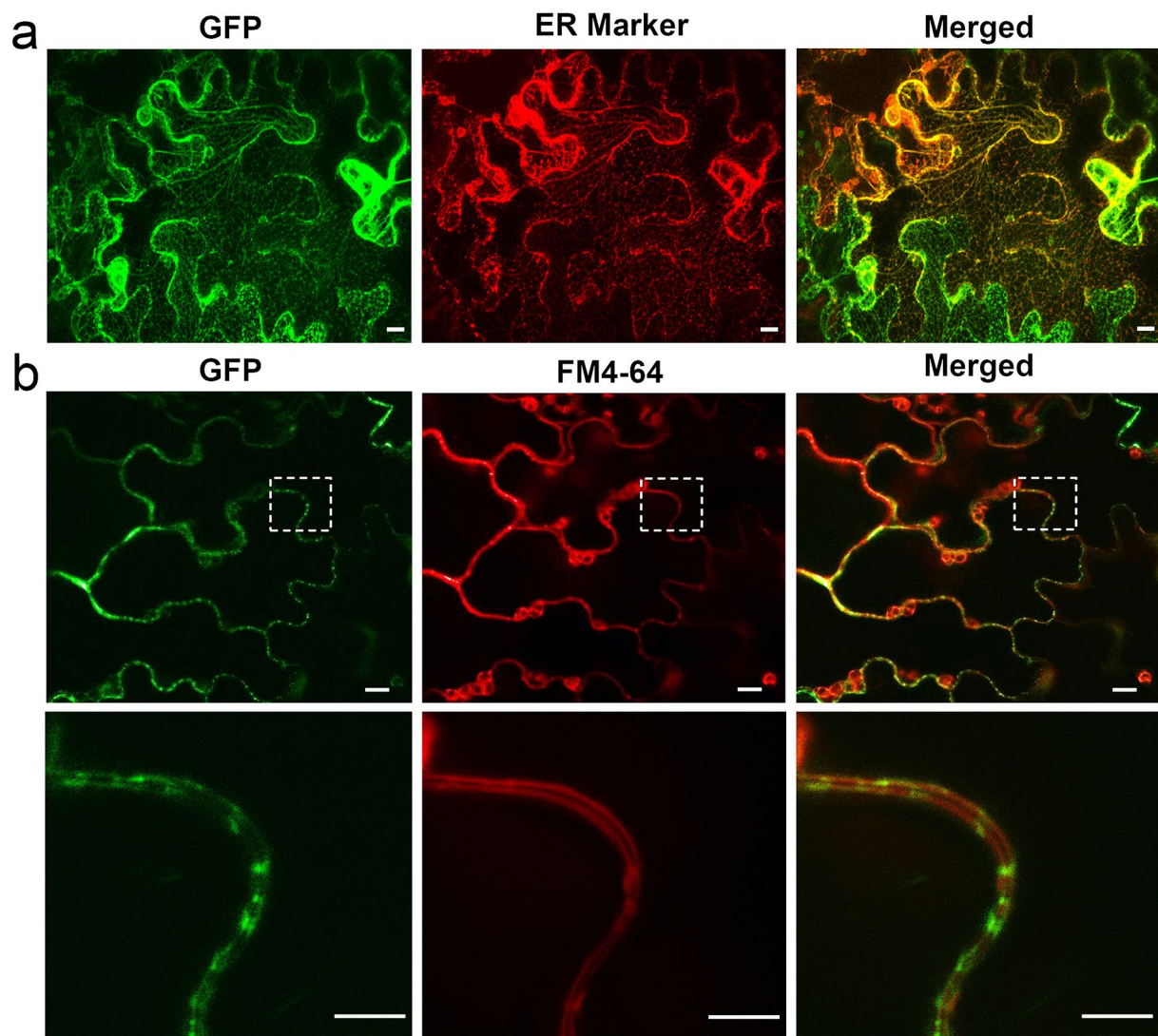


Fig. 2 Subcellular localization of SbYS1. The subcellular localization of SbYS1 in endoplasmic reticulum (a) and plasma membrane (b). Scale bars = 10 μ m, except for the three figures in the bottom (scale bars = 5 μ m) which are enlargements of boxes in the middle figures

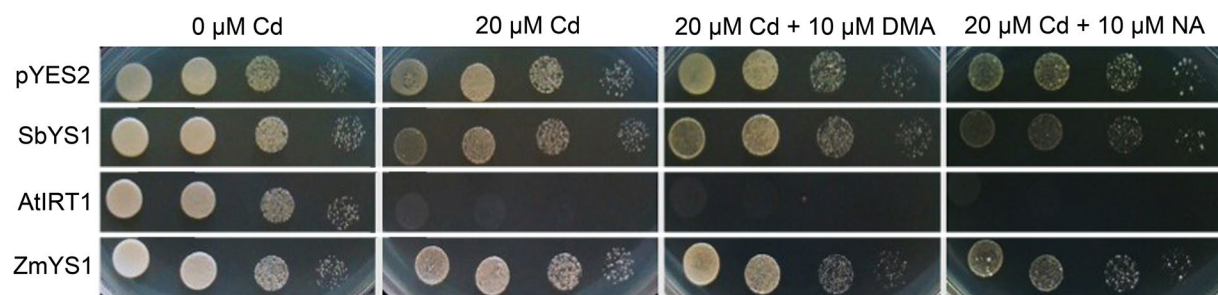


Fig. 3 Functional analysis of SbYS1 in yeast. pYES2, *SbYS1*, *AtIRT1* and *ZmYS1* represent yeast strains transformed with the empty vector pYES2, *SbYS1*, *AtIRT1* and *ZmYS1* genes, respectively. OD₆₀₀ values

of every yeast strain are 1, 0.1, 0.01 and 0.001 from left to right in different conditions

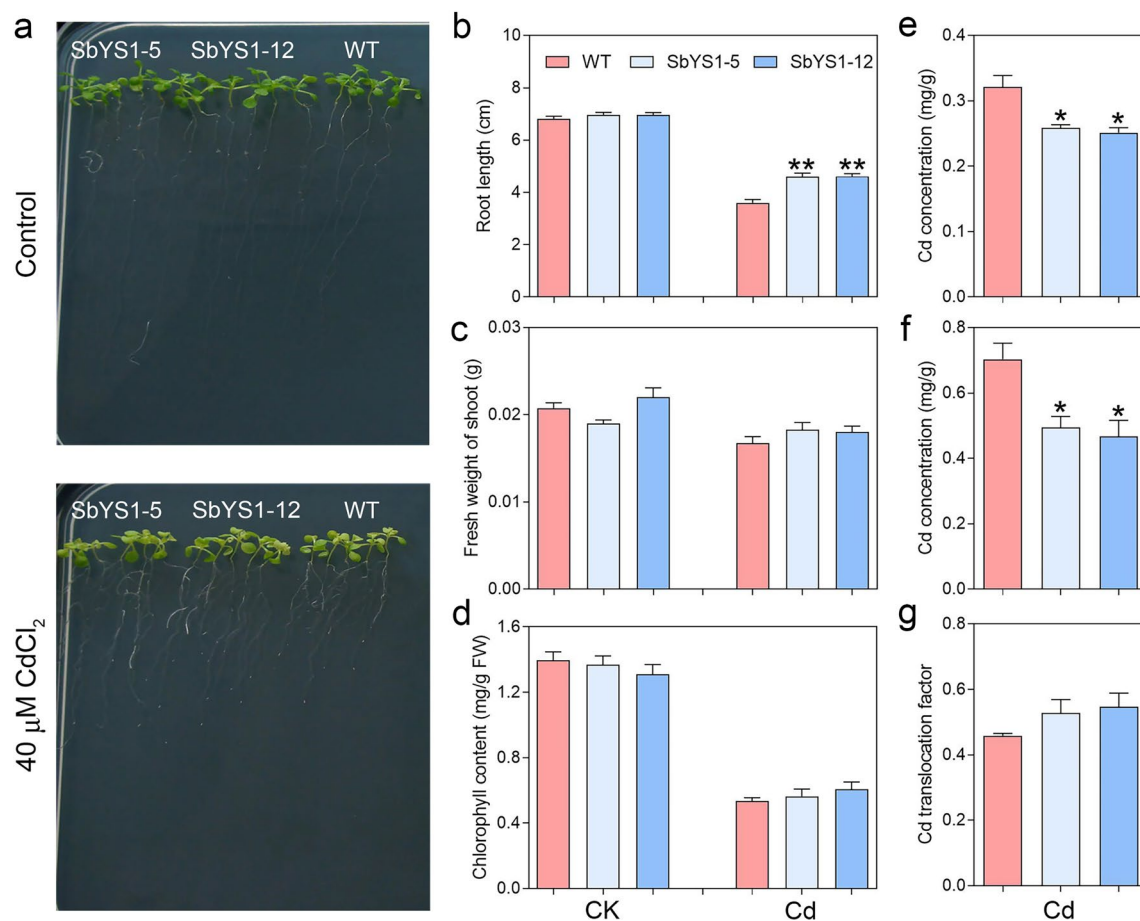


Fig. 4 Phenotypic characterization of *Arabidopsis* overexpressing *SbYS1* under cadmium stress. **a** The performance of *Arabidopsis* overexpressing *SbYS1* under control and 40 μM CdCl_2 conditions. **b** The root length ($n=20$). **c** Fresh weight of shoots ($n=4$). **d** Chlorophyll content ($n=5$). **e** Cd concentration in shoots ($n=3$). **f** Cd con-

centration in roots ($n=3$). **g** Cd translocation factor ($n=3$). Twenty plants for each replicate for **c-g**. Values are means $\pm$ SE. * and ** indicate significant difference at $P<0.05$ and $P<0.01$ by Student's *t*-test, respectively

transcription activity assay suggested *SbWRKY72* had no transcriptional activation activity and negatively regulated downstream target genes (Fig. 6b), which was consistent with the results of previous luciferase experiment described above.

In addition, the transcript level of *SbWRKY72* was examined in sweet sorghum by RT-qPCR. The results showed that *SbWRKY72* mainly expressed in roots. With the extension of treatment time, the expression level of *SbWRKY72* increased first and then decreased in roots as well as in shoots (Fig. 6c). Further Western-blotting analysis exhibited similar results that the expression level of *SbWRKY72* was induced by cadmium and mainly existed in roots (Fig. 6c).

Overexpression of *SbWRKY72* in *Arabidopsis* increased Cd accumulation in roots

The phenotype of *SbWRKY72* transgenic *Arabidopsis* showed no significant difference compared to wild-type

under control condition (Fig. 7a and b–d). When exposed to 75 μM CdCl_2 for one week, transgenic *Arabidopsis* were more sensitive to cadmium stress with decreased fresh weight and root length than that of wild-type, although no significant difference was found in chlorophyll content (Fig. 7a and b–d). Additionally, cadmium concentration in shoots was similar between wild-type and transgenic plants (Fig. 7e). However, more cadmium was accumulated in roots of transgenic plants (Fig. 7f), and overexpression of *SbWRKY72* in *Arabidopsis* decreased Cd translocation from roots to shoots (Fig. 7g).

Discussion

Phytosiderophores (PSs), such as nicotinic acid, mugenic acid and avenic acid, belong to a group of organic substances that generated by plants (Ueno et al. 2007). Under

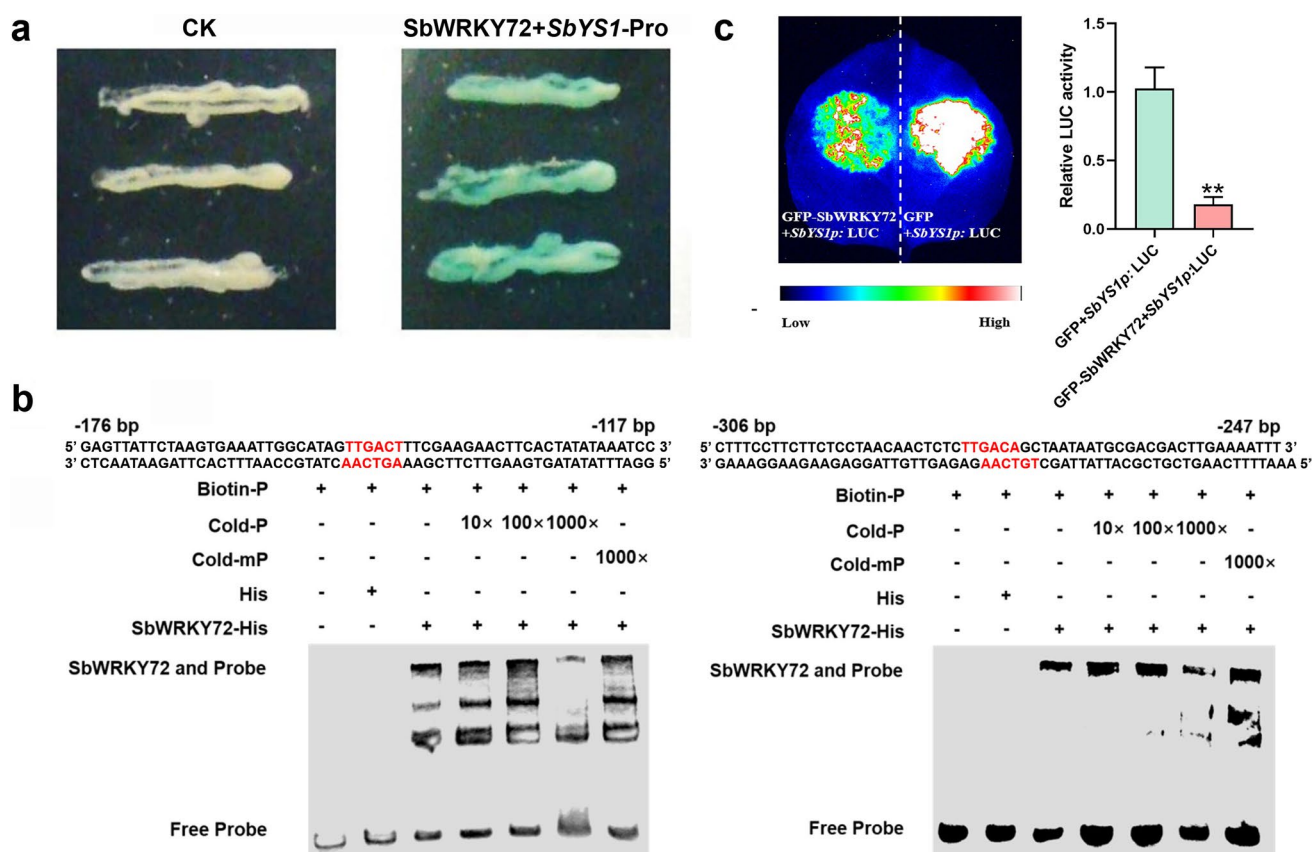


Fig. 5 Interaction of SbWRKY72 with the promoter regions of SbYS1. **a** Yeast one-hybrid assay. **b** Electrophoretic mobility shift assay. **c** Dual luciferase reporter assay, the left is a representative leaf

image, and the right is the quantification of corresponding relative luminescence intensities ($n=15$ independent leaves). **indicates a significant difference at $P<0.01$ by Student's t test

iron-deficient conditions, many graminaceous plants can secrete phytosiderophores in the rhizosphere to chelate insoluble iron and these chelates are further transported specially by members of the YSL family (Takagi 1976; Takagi et al. 1984; Römhelt and Marschner 1986; Schaaf et al. 2004). However, several transporters from the YSL family have been found to be involved in the uptake and translocation of cadmium, such as BjYSL7 from *Brassica juncea* and SnYSL3 from *Solanum nigrum* (Wang et al. 2013; Feng et al. 2017). In this study, we identified *SbYS1*, one member of the YSL family from sweet sorghum, was directly regulated by SbWRKY72 and affected cadmium accumulation and tolerance of plants.

The expression pattern of *SbYS1* indicated its function in Cd transportation

Under cadmium stress, *SbYS1* was mainly expressed in roots (Fig. 1a) and cross-sections observation based on GUS histochemical staining exhibited that *SbYS1* was mainly expressed in the epidermis and parenchyma of roots as well as stems, and in the vascular bundles and palisade tissues of

leaves (Fig. 1b–d). Feng et al. (2017) showed that *SnYSL3* from *Solanum nigrum*, which can transport nicotianamine complexes with Fe(II), Cu, Zn and Cd, was mainly expressed in epidermal cells and vascular tissues of roots and stems as well as in the vasculature of leaves under cadmium stress, implying that *SnYSL3* may function in the allocation of metal-NA into various organs of *S. nigrum* (Feng et al. 2017). In our study, *SbYS1* expressing in the epidermis of the roots indicated that they might be involved in the absorption process of metal ions. The expression in parenchyma tissues of roots and stems, and leaf vascular bundles indicated that they might play important roles in the transportation and distribution of metal ions.

For subcellular localization, lots of YSL family transporters have been reported to be located in plasma membranes, such as AtYS1, ZmYS1, HvYS1, OsYSL15 and OsYSL16 (Murata et al. 2006; Inoue et al. 2009; Ueno et al. 2009; Chu et al. 2010; Zheng et al. 2012). Nevertheless, some members from the YSL family exhibited different subcellular localizations. For example, HvYSL5 from barley was localized on the tonoplast (Zheng et al. 2011). AtYSL4 and AtYSL6 from Arabidopsis were localized

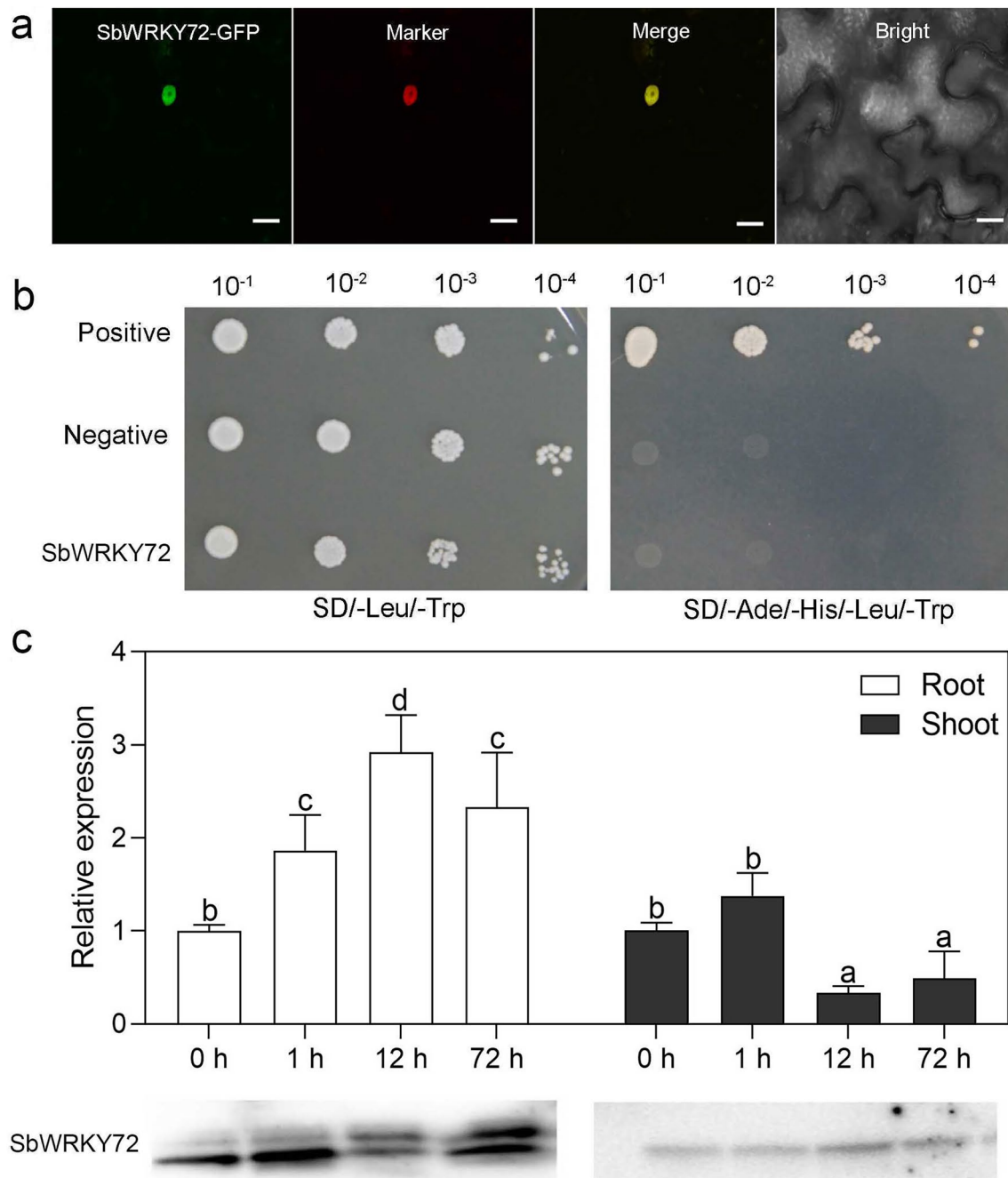


Fig. 6 Subcellular localization, transcriptional activity and expression pattern of SbWRKY72. **a** Subcellular localization of SbWRKY72, scale bars=20 μ m. **b** Transcriptional activation analysis of SbWRKY72. **c** Relative expression of SbWRKY72 at the transcript

(top) and protein (bottom) levels in roots and shoots of sweet sorghum which had been exposed to cadmium stress (10 μ M CdCl₂) for 0, 1, 12 and 72 h

to endomembrane system, including vacuole membranes and organelles like endoplasmic reticulum (Conte et al. 2013). Our results showed that SbYS1 was localized not only at the plasma membrane but also in the endoplasmic

reticulum (Fig. 2). Localization in the plasma membrane suggests that *SbYS1* may play a role in the absorption and transport of metal elements, endoplasmic reticulum

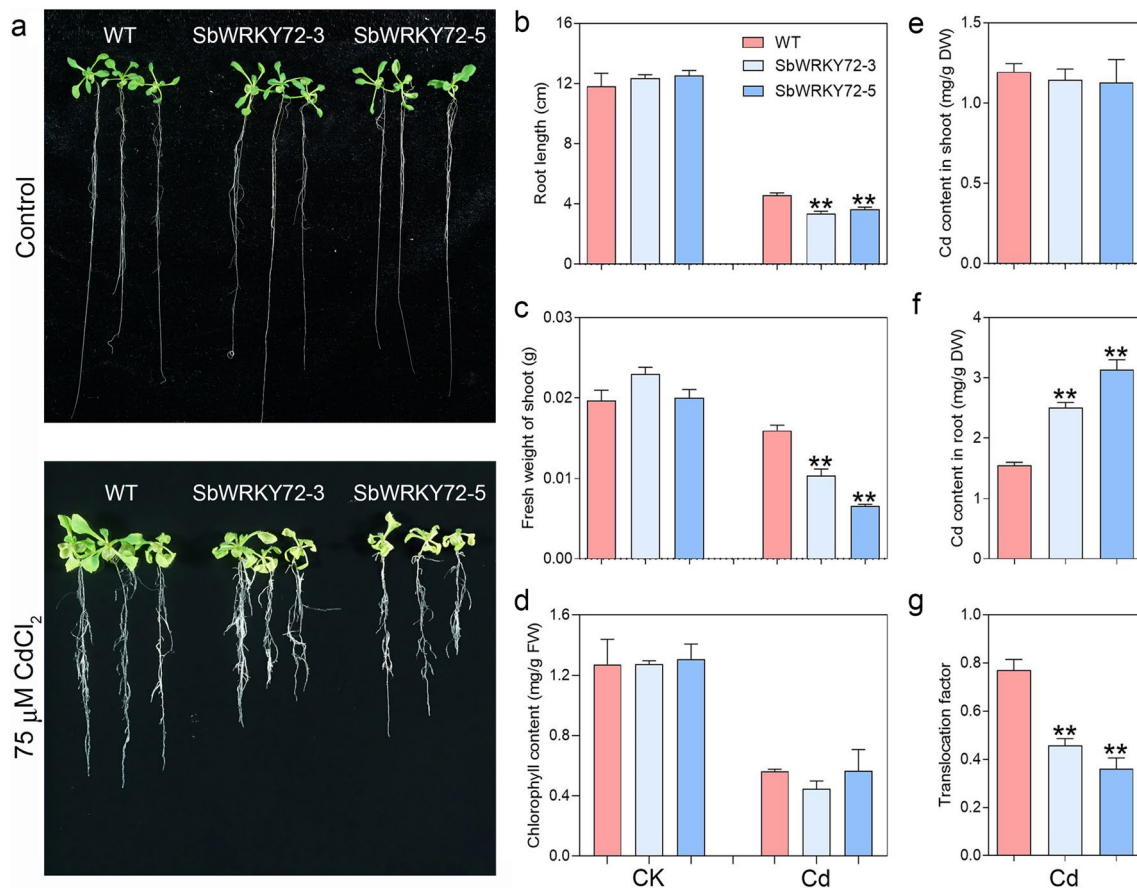


Fig. 7 Phenotype of Arabidopsis overexpressing *SbWRKY72* under cadmium stress. **a** Growth phenotype of *SbWRKY72*-overexpressing lines grown on control and 75 μM CdCl_2 conditions. **b** The root length ($n=3-4$). **c** Fresh weight of shoots ($n=4-6$). **d** Chlorophyll

content ($n=3$). **e** Cd concentration in shoots ($n=3$). **f** Cd concentration in roots ($n=3$). **g** Cd translocation factor ($n=3$). Twenty plants for each replicate for **c-g**. Values are means $\pm$ SE. ** indicates a significant difference at $P < 0.01$ by Student's t test

localization indicates that it may be responsible for providing metals to metalloproteins (Conte et al. 2013).

Overexpression of *SbYS1* in Arabidopsis enhanced its tolerance to cadmium

Members of the YSL family were identified to transport chelates of PS or its analogs with metals (Fe(III), Ni, Cu, Zn, Mn and Cd) (Schaaf et al. 2004; Chu et al. 2010; Zheng et al. 2012; Banakar et al. 2017). Our results suggested that *SbYS1* may have the ability to absorb cadmium in the form of Cd-NA but it is not sure whether *SbYS1* can absorb Cd-DMA (Fig. 3) although Schaaf et al. (2004) reported small currents provoked by Cd-DMA in *ZmYS1*-expressing oocytes, indicating that *ZmYS1* has the potential to take up Cd-DMA. Meda et al. (2007) reported that no obvious difference was observed between yeasts overexpressing *ZmYS1* and wild-type in the presence of 10 μM Cd plus 100 μM DMA. Therefore, they assumed that it is not the selectivity of the metal-phytosiderophore transporter *ZmYS1* that

restricted Cd-DMA uptake by *ZmYS1* (Schaaf et al. 2004) but the weak affinity of DMA to Cd. It is similar to our results that yeasts transformed with *SbYS1* and *ZmYS1* showed no significant difference compared to controls in the presence of 20 μM CdCl_2 plus 10 μM DMA (Fig. 3), which might also be related to the weak affinity of DMA to Cd.

Unexpectedly, overexpression of *SbYS1* in Arabidopsis increased root length and decreased Cd concentration in roots and shoots (Fig. 4). *MsYSL1* from *Miscanthus sacchariflorus*, *ZmYS1* from maize shared high similarity (95.87% for *MsYSL1* and 92.38% for *ZmYS1*) with *SbYS1* at the amino acid sequence level (Fig. S1). For *MsYSL1*, functional complementation experiments in yeast showed that it can transport Fe(II)-NA and Zn-NA, but not Cd-NA (Chen et al. 2018). Transgenic *MsYSL1* Arabidopsis exhibited a similar phenotype as *SbYS1* with increased root length and decreased Cd concentration in roots and shoots (Chen et al. 2018). The difference in transport substrate between *SbYS1* and *MsYSL1* may be due to their difference in sequence, especially the different regions between

the 317th and 389th (Fig. S1). Harada et al. (2007) reported that the 317th and 389th region of ZmYS1, which is the highly homologous protein of SbYS1 and MsYSL1, was associated with its substrate specificity. For ZmYS1, electrophysiological experiments in oocytes demonstrated that ZmYS1 can transport chelates of Fe(III), Ni, Zn, Cu, Mn and Cd with DMA, but whether ZmYS1 transport Cd-NA is not clear (Schaaf et al. 2004). *ZmYS1*-overexpressing *Arabidopsis* presented enhanced tolerance to cadmium stress when Fe-DMA was taken as a Fe source (Meda et al. 2007). SnYSL3 from *Solanum nigrum*, which shared 60% similarity with SbYS1 at the amino acid sequence level, can transport Cd-NA (Feng et al. 2017). Overexpression of *SnYSL3* in *Arabidopsis* decreased Cd concentration in roots and increased translocation ratio of Cd (Feng et al. 2017). These findings are similar to our results. In addition, our experiment showed that the expression of two genes for Cd exclusion (*AtDTX1* and *AtPDR8*) in transgenic *SbYS1* lines was higher than in the wild-type (Fig. S4), which is similar to the results in transgenic *MsYSL1* *Arabidopsis* (Chen et al. 2018). It is inferred that the activation of Cd efflux genes under cadmium stress may be partly associated with enhanced Cd tolerance and low Cd accumulation in shoots and roots of transgenic *SbYS1* *Arabidopsis*.

SbWRKY72 mediated Cd tolerance and accumulation in plants

Transcription factors of the WRKY family play important roles in plants resisting biotic and abiotic stresses (Fukushima et al. 2016; Yang et al. 2016; Lee et al. 2018). WRKY family members can act as a transcriptional activator of antioxidant enzymes for Cd tolerance. For example, Hong et al. (2017) reported that *ZmWRKY4* functioned in maize response to cadmium stress, which may activate maize antioxidant defense by regulating the expression of *ZmSOD4* and *ZmAPX*. On the other hand, WRKY transcription factors were involved in Cd tolerance and accumulation by regulating the expression of ion transporters. Sheng et al. (2019) reported that *AtWRKY13* positively regulated the expression of an ABC transporter PDR8, a Cd²⁺ extrusion pump, to mediate Cd tolerance and accumulation in *Arabidopsis*. Yang et al. (2016) found that *ThWRKY7* may regulate the expression of *ThVHAc1*, a V-ATPase c subunit (VHAc) of a proton channel, to participate in *Tamarix hispida* tolerance to cadmium stress. Our results showed that *SbWRKY72* negatively regulated the expression of metal transporter gene *SbYS1* and *Arabidopsis* overexpressing *SbWRKY72* became more sensitive to cadmium stress with a higher cadmium accumulation in roots (Figs. 5 and 7), which indicated that *SbWRKY72* may be involved in Cd tolerance and accumulation in sweet sorghum through regulating the expression of metal transporter gene *SbYS1*.

In summary, *SbWRKY72* can be considered as a candidate gene for improving the phytoremediation capacity of sweet sorghum in the future.

Conclusion

In this study, we identified a YSL transporter gene from sweet sorghum, *SbYS1*, which is induced by Cd stress and involved in Cd tolerance and accumulation of plants. Overexpression of *SbYS1* conferred Cd tolerance and decreased cadmium accumulation in roots and shoots simultaneously. *SbWRKY72* acted as a transcriptional inhibitor of *SbYS1*, transgenic *Arabidopsis* of *SbWRKY72* were more sensitive to Cd stress and accumulated more Cd in roots. The identification of *SbYS1* and its upstream transcription factor *SbWRKY72* provides new insights into the molecular regulation mechanisms of Cd tolerance and accumulation in sweet sorghum. These genes are of value for developing sweet sorghum with high Cd tolerance or accumulation ability through genetic manipulation, such as gene overexpression and knockout (or silencing), which can be used in the phytoremediation of Cd-contaminated soils.

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Author contributions LY: conceived and supervised this research, and revised the manuscript. JW: wrote the original draft, JW and GZ: performed experiments and analyzed data. LK and LS: helped revise and check the manuscript. All authors read and approved the final manuscript.

Data availability All the data generated or analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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